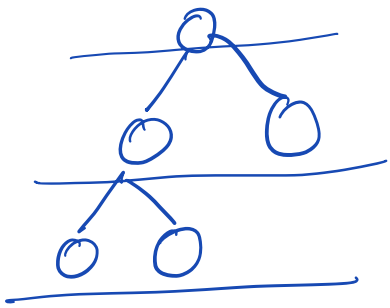
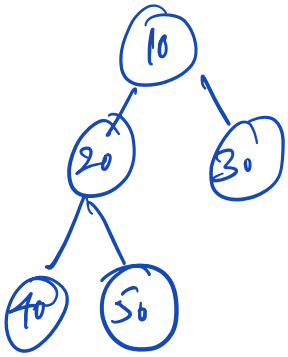
Heap is <u>complete binary tree</u>



① completely filled except last level.

② In last level, it is completely filled from left hand side.

10	20	30	40	50



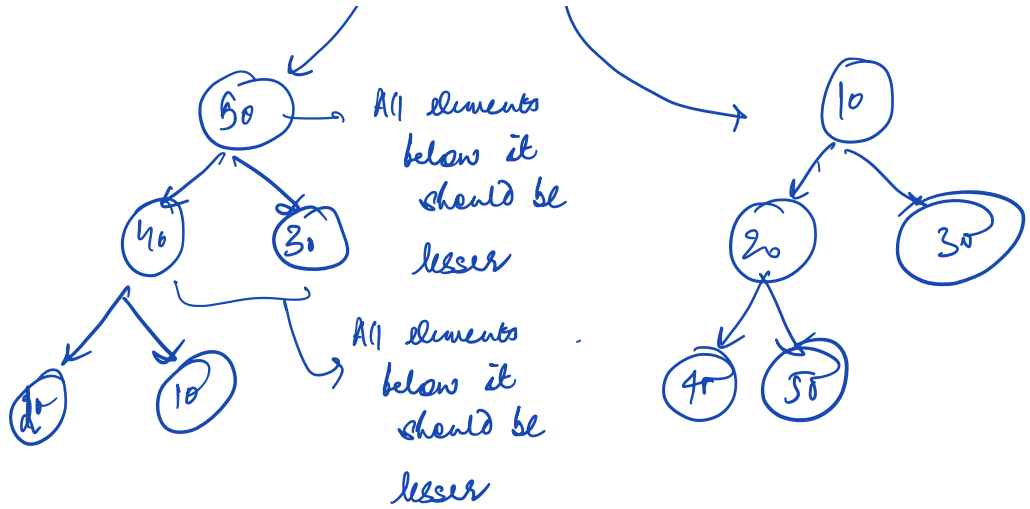
There can be many binary trees.

Heap → ① CBT

→ ② Higher priority element should come above a lower priority element.

Max heap → Max^m element has higher priority

Min Heap → Min^m element has higher priority.



Why do we need heap

a = [ ]

Q → N operations

1) Add any element into the list.

2) Report the max element in the list at that point of time.

4, 2, 6, 3, 8

<1,4>

<1,2>

<1,6>

<1,3>

<2 → 6

Report operation (max)

<1,8>

2 → 8

Sorted Array

[2,3,4,10,20,6]

Insertion → O(N)

Max element → O(1)

eg Number of Queries, $N = 10^5$

Thus, overall complexity becomes $O(N^2)$

1) Add elements → Thus it is $O(\log N)$

2) Max value → max value → $O(1)$

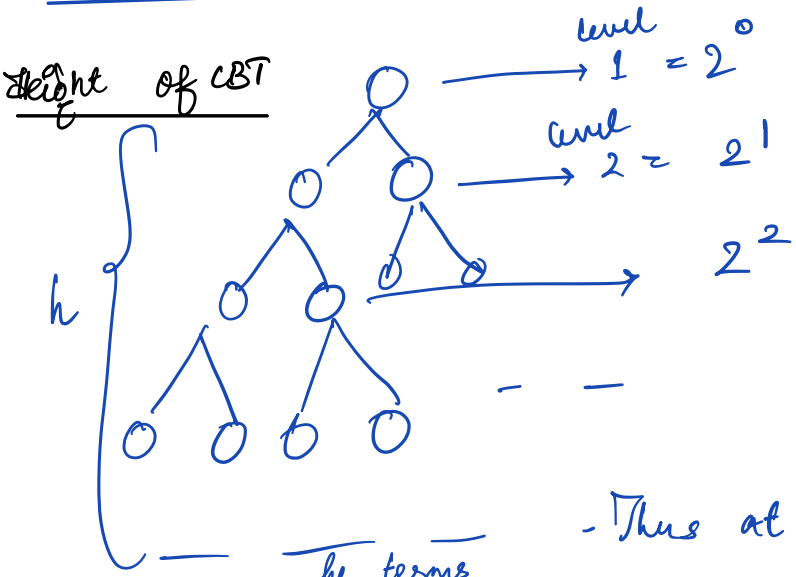
Max Heap → Root of the heap

Thus $O(N^2)$ → reduced to $N \log N$.

* Heaps are primarily used for effecient access to min^m (min-heap) or max^m (max-heap) element usually in $O(1)$ time.

→ Heaps are not directly used for graph transversal algorithms. They can be used in algorithms like Dijkstra's shortest path to effeciently access smallest node.

Height of CBT



- Thus at level h → 2^{h-1} nodes

$$\underbrace{1 + 2^1 + 2^2 + \dots + 2^{h-1}}_{\text{GP}} \rightarrow \text{No. of Nodes}$$

h terms

$$\rightarrow \frac{2^h - 1}{2 - 1} = N$$

$$\text{Sum of } n \text{ terms} = \frac{[a(1-r^n)]}{1-r}$$

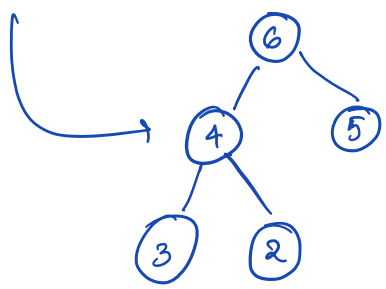
$$2^h = N+1$$

$$\boxed{h_{CBT} = \log_2(N+1)}$$

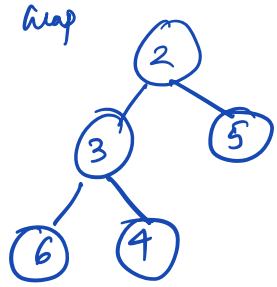
Element stored in heap and heap order

Max heap

4 6 5 3 2

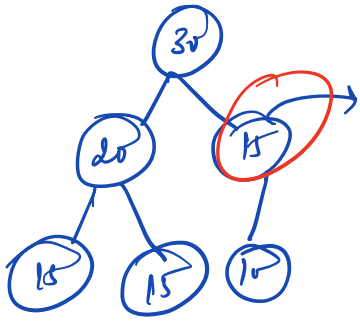


Min heap



Why its not stable?

20	30	10	15	15	15



This 15 comes before. Thus, order is not proper. Thus, heap is not stable.

How to insert in a heap?